

SCENARIO BASED ANALYSISQ1. MORPHOLOGY PROCESSING FOR SEARCH ENGINE :

A Search engine receives different forms of same words such as analyzing, analysis and analytical. If these words are stored separately, the search engine may miss relevant documents. Therefore, all related words must be normalized to a common base form.

INPUT WORDS :

- Analyzing
- analysis
- analytical

MORPHOLOGICAL ANALYSIS

WORD	ROOT	AFFIX	TYPE	NORMALIZED
analysis	analyze	sis	Derivational	analysis
analyzing	analyze	ing	Inflectional	analyze
analytical	analyze	tical	Derivational	analyze

OUTPUT :

All three words are normalized to analyze to improve search accuracy.

Q2) MORPHOLOGICAL PARSING FOR SENTIMENT ANALYSIS

A sentiment analysis system receives word like disagree, agreement and agreeable. The system must identify the base word and understand how prefixes and suffixes changes the meaning.

INPUT WORDS :

- Disagree
- Dis agreement
- agreeable

MORPHOLOGICAL ANALYSIS :

WORD	PREFIX	ROOT	SUFFIX	TYPE	MEANING
Disagree	dis	agree	-	Derivational	-ve meaning
agreement	-	agree	ment	Derivational	Noun Form
agreeable	-	agree	able	Derivational	Adjective Form

OUTPUT :

All words are normalized to agree for consistent sentimental analysis.

Q3) MORPHOLOGY--BASED NORMALIZATION:-

A news analytical system process govern - government and governance. The system should map all word forms to the same lexical representation.

INPUT WORDS:-

- Govern
- Government
- Governance

MORPHOLOGICAL ANALYSIS:-

WORD	ROOT	APPR	Type	Normalized
govern	govern	-	Base	govern
government	govern	ment	Derivational	govern
governance	govern	ance	Derivational	govern

OUTPUT:-

All words are normalized to the govern for document clustering.

4. MORPHOLOGICAL PASSING:-

A document classification system process activate, activation, and reactivation. It must identify prefix, suffix, and the root word to improve semantic grouping.

INPUT WORDS:-

- Active
- Activation
- Reactivation

MORPHOLOGICAL ANALYSIS:-

WORD	PREFIX	ROOT	SUFFIX	TYPE
active	-	activate	-	Base Form
activation	-	activate	ion	Derivational
reactivation	re	activate	ion	Derivational

OUTPUT:-

All words are normalized to active

5. INFLECTIONAL MORPHOLOGY:-

A search optimization module process create, creators, and creating. The system removes grammatical variations and stores the common base words.

INPUT WORDS:-

- Create
- Creators
- Creating

MORPHOLOGICAL ANALYSIS:-

WORD	ROOT	SUFFIX	GRAMMER	TYPE	NORMALIZED
Create	Create	-	Base form	Base	Create
Creators	Create	s	Third Person singular	Inflectional	create
Creating	Create	ing	Present participle	Inflectional	create

OUTPUT:-

All words are normalized to create for efficient search and informational retrieval.